

GEOLOGY • UPPER SIXTH

UPPER SIXTH — GEOLOGY

CHAPTER 8

ENVIRONMENTAL GEOLOGY

SUGGESTED TIME: 7 PERIODS

IN THIS CHAPTER

This chapter explores the interaction between geological processes and human activities, covering natural hazards, their mitigation, climate change impacts, and pollution. It equips students with the analytical tools to assess environmental risks and propose sustainable solutions.

- Define environmental geology and explain its relevance to sustainable development.
- Identify and classify common natural geologic hazards in Cameroon and worldwide.
- Analyse the primary and secondary effects of earthquakes, landslides, volcanic eruptions, and floods.
- Design and evaluate mitigation strategies for each major geohazard.
- Explain the mechanisms of global warming and its influence on climate patterns.
- Assess the sources, pathways, and impacts of major pollutants on soil, water, and air.



DOWNLOAD ON- BUCH

Free • Google Play
AI tutor, past papers,
results



GO ONBUCH+

All signed courses, unlimited AI
tutor, PDF sheets — OnBuch+
tab in the app

THIS SIGNED COURSE CONTAINS

COURSE

complete, illustrated,
step by step

METHODS

and worked examples

EXERCISES

exam style + solutions

SUMMARY SHEET

the essentials to
remember



Signed • OnBuch+

ONBUCH+ TEACHING TEAM
FOLLOWS THE OFFICIAL MINESEC SYLLABUS

OnBuch+

CONTENTS

1. Foundations of Environmental Geology and Major Geohazards
2. Effects and Mitigation of Geohazards – Part 1
3. Effects and Mitigation of Geohazards – Part 2
4. Global Warming and Its Effects on Climate
5. Pollution – Sources, Transport, and Management
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- Define environmental geology and explain its relevance to sustainable development.
- Identify and classify common natural geologic hazards in Cameroon and worldwide.
- Analyse the primary and secondary effects of earthquakes, landslides, volcanic eruptions, and floods.
- Design and evaluate mitigation strategies for each major geohazard.
- Explain the mechanisms of global warming and its influence on climate patterns.
- Assess the sources, pathways, and impacts of major pollutants on soil, water, and air.
- Integrate knowledge of hazards, climate change, and pollution in a comprehensive environmental impact assessment.

PREREQUISITES

- Basic plate tectonics and rock cycle (Lower Sixth).
- Fundamentals of weathering, erosion, and sediment transport.
- Understanding of the greenhouse effect and carbon cycle.
- Basic chemistry of common pollutants (heavy metals, hydrocarbons).

1 FOUNDATIONS OF ENVIRONMENTAL GEOLOGY AND MAJOR GEOHAZARDS

In the bustling streets of Douala, sudden flash floods turn markets into rivers within a few hours of torrential rain, while the dark volcanic slopes of Mount Cameroon remind the villages of Buea, Limbe and Idenau that eruptions have repeatedly sent lava flows down towards the sea. These everyday scenes are not accidents of fate: they are the visible expression of geological processes that shape our environment and, when they intersect with human settlement, threaten communities. The central problem of this chapter is therefore the following: how can geological knowledge be used to anticipate, reduce and manage the dangers that the Earth itself poses to people, property and ecosystems? Answering that question is precisely the mission of environmental geology.

1.1 Definition and Scope of Environmental Geology

DEFINITION Environmental Geology

Environmental geology is the application of geological principles, data and methods to the identification, understanding and solution of environmental problems, so as to minimise the degradation of the environment and the loss of human life and property caused by the interaction between geological processes and human activities.

The definition contains two essential ideas that candidates must be able to unpack in an examination.

- **Geology as a tool.** Environmental geology is not a new science separate from geology; it uses the classical tools of the geologist (field mapping, stratigraphy, petrology, structural analysis, geophysics, hydrogeology) but directs them towards practical problems: where to build, where to drill a well, where to dump waste, where a slope is unstable.
- **Interaction between nature and society.** A lava flow in an uninhabited forest is a geological process; the same flow crossing a road and destroying houses is an environmental problem. Environmental geology studies precisely this interface.

The **scope** of the discipline is broad. It covers, among others:

1. the study and mitigation of **natural geologic hazards** (earthquakes, volcanic eruptions, landslides, floods);
2. the management of **earth resources** (water, minerals, soils, energy) and the environmental consequences of their exploitation;
3. the study of **land use**, including the choice of sites for settlements, dams, roads and waste-disposal facilities;
4. the study of **environmental pollution** of soil, water and air, and of global phenomena such as climate change.

DID YOU KNOW? Environmental geology in Cameroon

Cameroon is sometimes called “Africa in miniature” for its geology too: an active volcano (Mount Cameroon), a seismically active volcanic line, unstable lateritic slopes in the humid south, flood-prone coastal lowlands around Douala, and the deadly gas-charged crater lakes of the Western Highlands (Nyos, Monoun). Few countries offer such a complete natural laboratory for environmental geology.

1.2 Interdisciplinary Links

No environmental problem is purely geological. The environmental geologist works at the cross-roads of several disciplines.

Partner discipline	Contribution to environmental geology
Geology	Provides the understanding of Earth materials, structures and processes (faults, magma, slopes, aquifers).
Ecology	Assesses how geological events and pollution affect living organisms and ecosystems (e.g. the suffocation of fauna around Lake Nyos).
Engineering	Designs mitigation structures: retaining walls, drainage systems, seismic-resistant buildings, lava-diversion barriers.
Policy and land-use planning	Translates scientific hazard maps into laws, building codes, evacuation plans and protected zones.

EXAMPLE The Lake Nyos degassing project

After the 1986 Lake Nyos disaster, geologists identified the cause (accumulation of magmatic CO₂ in deep lake water), engineers designed and installed degassing pipes that siphon gas-rich bottom water to the surface, ecologists monitored the recovery of the area, and the Cameroonian government organised the resettlement of survivors and controlled access to the lake. This is environmental geology in action: four disciplines cooperating on one problem.

1.3 Classification of Natural Geologic Hazards

DEFINITION Geologic hazard

A **geologic hazard** (or **geohazard**) is a natural geological process or event that has the potential to cause damage to human life, property or the environment.

Geohazards are classified according to the geological process involved into four main cate-

gories.

1. Seismic hazards.

These result from sudden movements of the Earth's crust along faults, generating **earthquakes**. Effects include ground shaking, surface rupture, liquefaction of water-saturated soils and tsunamis. In Cameroon, the Cameroon Volcanic Line is associated with moderate seismicity, felt in the Mount Cameroon region and along the oceanic segment.

2. Volcanic hazards.

These are linked to the rise and eruption of **magma**: lava flows, ash falls, pyroclastic flows, volcanic gases and lahars (volcanic mudflows). Mount Cameroon, one of Africa's most active volcanoes, erupted several times in the 20th century (notably in 1922, 1954, 1959, 1982, 1999 and 2000), with lava flows reaching the sea near Idenau in 1999.

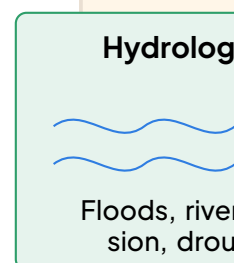
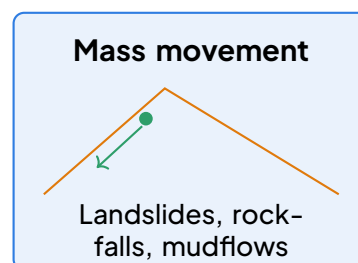
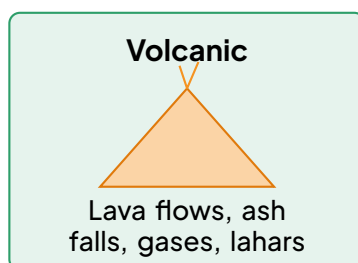
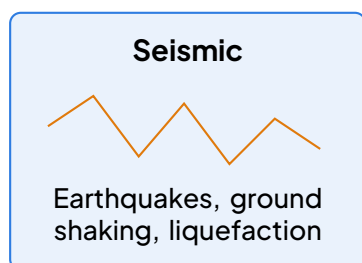
3. Mass movement hazards.

These involve the downslope movement of rock, soil or debris under gravity: **landslides**, rock-falls, mudflows and slumps. They are favoured by steep slopes, deeply weathered (lateritic) rocks, heavy rainfall and deforestation. The wet, hilly regions of the North-West, South-West and West Regions of Cameroon are particularly prone to them.

4. Hydrological hazards.

These concern the behaviour of surface and ground water: **floods**, riverbank erosion, and droughts related to falling water tables. Flash floods in Douala, where intense rainfall meets a low-lying, densely built coastal plain with inadequate drainage, are a recurrent example.

The four categories of natural geologic hazards



Classification of natural geologic hazards into four categories, with the main processes in each.

WARNING Hazard is not risk

A very common mistake is to confuse **hazard** with **risk**. The hazard is the natural process itself (e.g. an eruption of Mount Cameroon); it exists whether or not people are present. The risk expresses the potential losses because people and property are exposed to the hazard. An eruption on an uninhabited slope is a hazard with almost zero risk; the same eruption above a crowded village is a high risk. Examiners regularly test this distinction.

1.4 From Hazard to Risk: Susceptibility, Vulnerability, Risk

To manage geohazards rationally, we must quantify them. Three quantities are distinguished.

DEFINITION Susceptibility, vulnerability and risk

Susceptibility is the propensity of an area to experience a given hazard, based on its physical characteristics (slope, rock type, proximity to a fault or volcano, rainfall). **Vulnerability** is the degree of damage that exposed elements (people, buildings, crops) would suffer if the event occurred. **Risk** combines the probability of the hazardous event with the severity of its consequences.

PROPERTY The risk equation

The risk associated with a geohazard can be expressed as

$$\text{Risk} = \text{Probability of occurrence} \times \text{Severity of consequences,}$$

often written $R = P \times C$, where P is the probability (or frequency) of the hazardous event in a given period and C measures the expected losses (deaths, cost of damage). The severity of consequences itself depends on the vulnerability of the exposed population and the value of the elements at risk. This relation is used qualitatively at GCE level; no formal proof is required, but candidates must be able to apply it in simple numerical situations.

WORKED EXAMPLE 1 Comparing two villages on the slopes of Mount Cameroon

Village A lies on the 1999 lava-flow path, 2 km from a recent eruptive fissure; geologists estimate the probability of a damaging lava flow there at 0.02 per year, and a flow would destroy most houses (consequence severity rated 0.9 on a scale from 0 to 1). Village B lies 15 km away on older terrain, with probability 0.002 per year and severity 0.3. Compare the risks.

SOLUTION

Apply the risk equation $R = P \times C$ to each village.

$$R_A = 0.02 \times 0.9 = 0.018 \text{ per year,}$$

$$R_B = 0.002 \times 0.3 = 0.0006 \text{ per year.}$$

The risk at village A is $\frac{0.018}{0.0006} = 30$ times greater than at village B. Conclusion: although eruptions may occur anywhere on the volcano, land-use planning should restrict dense settlement near recent flow paths, because both the probability and the severity are highest there.

1.5 Case Studies from Cameroon

1. The eruptions of Mount Cameroon.

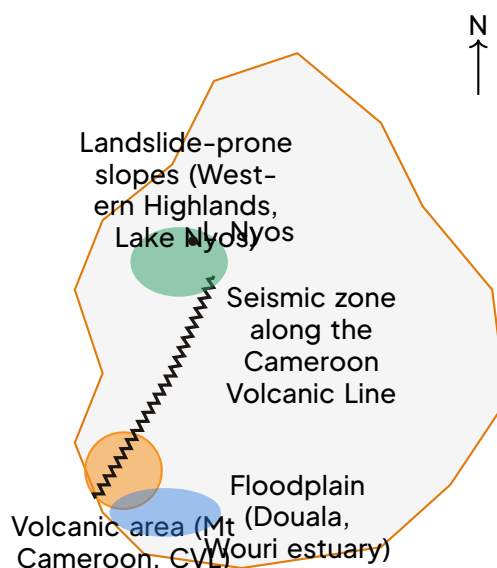
Mount Cameroon (4095 m) is the most active volcano in West and Central Africa. Historical eruptions (1922, 1954, 1959, 1982, 1999, 2000) produced basaltic lava flows from flank fissures. In 1999, two flows descended the south-west flank; one crossed the Limbe–Idenau road and reached the Atlantic Ocean, cutting communications and threatening plantations. The hazards include lava flows, ash fall on crops and roofs, and toxic gases. Because the fertile volcanic soils attract dense settlement and plantations, vulnerability is high.

2. The 1986 Lake Nyos gas burst.

On 21 August 1986, a massive cloud of carbon dioxide, CO₂, was suddenly released from the deep waters of Lake Nyos, a crater lake in the North-West Region. Being denser than air, the gas flowed down valleys and suffocated about 1700 people and thousands of livestock up to several kilometres from the lake. The gas is of magmatic origin, dissolving slowly into the cold bottom water until the lake became saturated and overturned (a limnic eruption). A similar event had killed 37 people at Lake Monoun in 1984. Mitigation now relies on degassing pipes that continuously bring gas-rich bottom water to the surface in a controlled fountain.

3. The Douala floods.

Douala lies on a low, flat coastal plain crossed by the Wouri and Dibamba rivers, with a hot, very wet climate. Intense rains, high tides, poor drainage and rapid unplanned urbanisation combine to produce recurrent flash floods; the severe floods of 2012 inundated whole neighbourhoods, markets and roads, displacing residents and damaging property. This is a hydrological hazard amplified by human vulnerability: blocked drains, construction on floodplains and removal of mangroves.



Schematic hazard map of Cameroon: volcanic area of the Cameroon Volcanic Line (orange), associated seismic zone (zigzag), landslide-prone slopes of the Western Highlands (green) and the Douala–Wouri floodplain (blue).

1.6 Hazard Mapping Basics

A **hazard map** is a map that shows the spatial distribution of a hazard, usually as zones of different susceptibility. Modern hazard maps are produced with a **Geographic Information System** (GIS), which superposes digital layers of information.

METHOD Producing a simple hazard susceptibility map with GIS layers

1. Define the hazard to be mapped (e.g. landslides) and the study area.
2. Collect the controlling layers: slope steepness (from a digital elevation model), rock type and degree of weathering (geological map), rainfall, land use and vegetation cover, and the location of past events (an inventory map).
3. Weight each layer according to its influence on the hazard (steep slopes and weathered rocks receive high weights).
4. Overlay the layers in the GIS and combine the weights to compute a susceptibility score for every point of the area.
5. Classify the scores into zones (low, moderate, high susceptibility) and assign colours.
6. Validate the map by checking that past events fall mostly inside the high-susceptibility zones, then present the map to planners.

KEY POINTS Key points of this section

- Environmental geology applies geological knowledge to solve environmental problems and protect communities.
- It is interdisciplinary, linking geology with ecology, engineering and public policy.
- Natural geohazards fall into four categories: seismic, volcanic, mass movement and hydrological.
- Hazard is the natural process; risk equals probability of occurrence multiplied by severity of consequences, and grows with human vulnerability.
- Cameroon offers textbook examples: Mount Cameroon eruptions, the Lake Nyos gas burst of 1986 and the Douala floods.
- Hazard susceptibility maps are built by overlaying weighted GIS layers and validated against past events.

EXERCISE 1 Hazard, vulnerability and risk

1. Define environmental geology and state two of its fields of application.
2. Distinguish clearly between a hazard and a risk, illustrating your answer with the Lake Nyos event.
3. A district on the flank of Mount Cameroon has an estimated probability of 0.01 per year of being affected by a lava flow, with a consequence severity of 0.8. Calculate the annual risk and explain what would happen to the risk if the population of the district doubled.
4. List the four categories of natural geologic hazards and give one Cameroonian example of each.

SOLUTION Exercise 1

1. Environmental geology is the application of geological principles to the identification and solution of environmental problems. Any two of: geohazard study and mitigation, management of earth resources, land-use planning and waste-disposal siting, study of pollution and climate change.
2. The hazard is the natural process: the accumulation and sudden release of magmatic CO_2 from Lake Nyos. The risk is the potential loss because people and livestock lived in the valleys downwind of the lake; the same gas burst in an uninhabited area would have been a hazard with negligible risk.
3. $R = P \times C = 0.01 \times 0.8 = 0.008$ per year. If the population doubled, the severity of consequences C (more people and property exposed) would increase, so the risk would increase even though the hazard (the probability P) is unchanged.
4. Seismic: earthquakes along the Cameroon Volcanic Line. Volcanic: eruptions of Mount Cameroon (e.g. 1999). Mass movement: landslides on the steep slopes of the Western Highlands. Hydrological: flash floods in Douala (e.g. 2012).

2 EFFECTS AND MITIGATION OF GEOHAZARDS – PART 1

In the previous section we identified the major geohazards that threaten human societies — earthquakes, volcanic eruptions, landslides, floods and coastal hazards — and we saw that a **hazard** only becomes a **disaster** when it strikes a vulnerable, exposed population. We now move from recognising hazards to living with them. This section focuses on the earthquake, the deadliest sudden-onset geohazard, and answers two questions that every GCE candidate must mas-

ter: what exactly does an earthquake do to the ground and to structures? and what can engineers and communities do to reduce the losses? Cameroon is directly concerned: the Mount Cameroon region and the Cameroon Volcanic Line are seismically active, and the 2022 seismic swarms felt around Buea reminded us that preparedness is not an abstract luxury.

2.1 Primary Effects of Earthquakes

Primary effects are the consequences produced directly by the passage of seismic waves and by fault movement itself. Three of them dominate: ground shaking, surface rupture and liquefaction.

Ground shaking.

When elastic strain energy accumulated along a fault is suddenly released, seismic waves radiate outward and set the ground in motion. The severity of shaking at a site, called the **intensity**, depends on the magnitude of the event, the distance from the focus, the depth of the focus, and — critically — the nature of the local ground. Soft, water-saturated sediments (such as the alluvial fills of the Wouri estuary around Douala) amplify shaking far more than solid bedrock, a phenomenon known as **site amplification**. Engineers quantify shaking by the **peak ground acceleration** (PGA), the maximum acceleration recorded at the surface, expressed in m s^{-2} or as a fraction of g ($g = 9.81 \text{ m s}^{-2}$).

PROPERTY The magnitude scale is logarithmic

The magnitude M of an earthquake (Richter M_L or moment magnitude M_w) is a logarithmic measure of wave amplitude (and hence of energy). Each increase of one magnitude unit corresponds to approximately 31.6 times more energy released:

$$\frac{E_2}{E_1} \approx 31.6^{M_2 - M_1}.$$

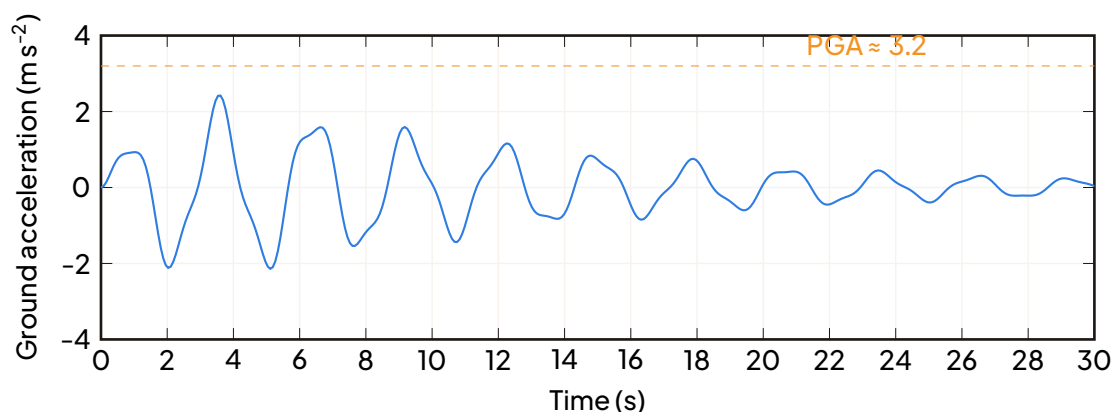
Thus a magnitude 7.0 event releases about $31.6 \times 31.6 \approx 1000$ times more energy than a magnitude 5.0 event. (The derivation from the definition of the scale is not examinable, but the factor 31.6 and its use in calculations are.)

EXAMPLE Comparing two earthquakes

How many times more energy does a magnitude 6.0 earthquake release than a magnitude 4.0 earthquake?

$$\frac{E_6}{E_4} = 31.6^{6-4} = 31.6^2 \approx 998.6 \approx 1000.$$

The magnitude 6.0 event releases roughly one thousand times more energy. This is why "only two points higher on the scale" is a dangerously misleading phrase.



Idealised accelerogram for a magnitude 6.0 event: acceleration rises rapidly to the peak ground acceleration (PGA), then decays as the seismic energy is dissipated.

Surface rupture.

If the fault break reaches the surface, the ground is physically offset — sometimes by several metres — along a line that can extend for tens of kilometres. Any structure, road, pipeline or water main built across the rupture is torn apart, and no realistic building design can resist a direct offset. The only mitigation is avoidance: mapping active fault traces and forbidding construction in a buffer zone on either side.

Liquefaction.

Perhaps the most insidious primary effect is liquefaction, which turns apparently solid ground into a fluid.

DEFINITION Liquefaction

Liquefaction is the loss of strength and stiffness of a saturated, loose, cohesionless soil (typically sand or silt) when earthquake shaking increases the pressure of the water in the pores between grains. The grains momentarily lose contact with one another, the soil behaves like a liquid, and it can no longer support foundations.

The intuition is simple: dry sand in a bucket supports your weight because the grains press against each other. Shake the bucket violently while it is full of water and the grains are pushed apart by pressurised pore water; the sand "boils", and a heavy object placed on it sinks. In the field, liquefaction produces sand boils (eruptions of sandy water at the surface), tilting and sinking of buildings, flotation of buried tanks and pipelines, and lateral spreading of river banks. Coastal and estuarine districts built on young, water-saturated sands — such as parts of Douala and Limbe — are the most susceptible in Cameroon.

2.2 Secondary Effects of Earthquakes

Secondary effects are triggered by the shaking rather than caused directly by the seismic waves. They frequently kill more people than the shaking itself.

- **Tsunamis.** A sudden vertical displacement of the sea floor during a submarine earthquake displaces the entire water column above it. In the open ocean the wave is long and low, but as it enters shallow coastal water it slows and grows in height (shoaling), arriving as a devastating surge. Cameroon has a low but non-zero exposure along the Gulf of Guinea.
- **Landslides.** Shaking destabilises steep slopes, especially where rocks are weathered or slopes are over-steepened by road cuts. The volcanic slopes of Mount Cameroon and the escarpments of the Western Highlands (e.g. around Bafoussam and Dschang) are prone to earthquake- and rain-triggered slides that bury roads and villages.
- **Fires.** Ruptured gas lines, overturned stoves and short-circuited electrical cables ignite fires at the very moment when water mains are broken and roads are blocked, so fire-fighting becomes nearly impossible. Fire following earthquake has historically destroyed more buildings than the shaking in several great disasters.
- **Disease outbreaks.** Damage to water supply and sanitation systems contaminates drinking water; crowded emergency shelters favour the spread of cholera, dysentery and respiratory infections. In Cameroon, where cholera is endemic in some regions, this secondary effect must be taken very seriously.

WARNING Do not ignore secondary hazards

A very common mistake — in examinations and in real planning — is to assess only the primary effect (shaking) and to forget the secondary effects. Ignoring secondary hazards leads to a systematic underestimation of total risk. A complete answer on earthquake effects must always mention at least two secondary effects and explain the chain of causation: shaking → slope failure / pipe rupture / sea-floor displacement → landslide / fire / tsunami → additional casualties and losses.

2.3 Seismic Building Codes and Retrofitting

Since we cannot prevent earthquakes, we reduce casualties by ensuring that buildings do not collapse. The guiding principle of seismic design is often summarised as: "earthquakes do not kill people; buildings do."

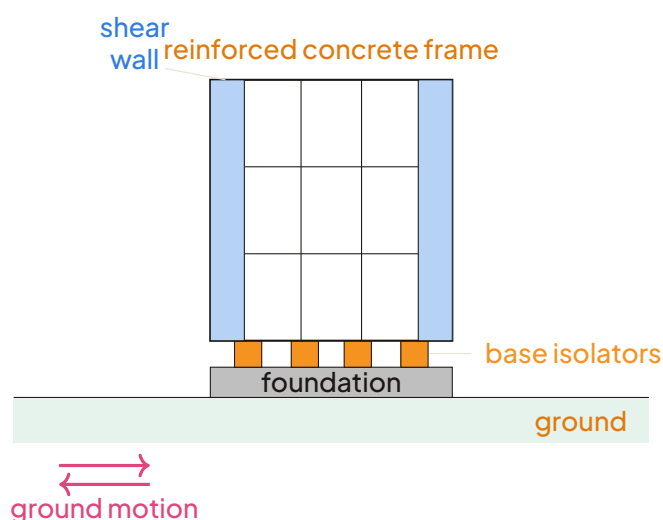
DEFINITION Seismic building code

A **seismic building code** is a set of legally enforced design rules that specifies the minimum horizontal forces a structure must resist, the materials and detailing to be used, and the construction quality controls required, so that a building may be damaged in a strong earthquake but must not collapse.

Key engineering techniques include:

- **Shear walls:** stiff reinforced-concrete or masonry walls, continuous from foundation to roof, that resist horizontal (lateral) forces and prevent the soft-storey collapse typical of open ground floors.

- **Base isolation:** flexible bearings (layers of rubber and steel, or sliding pads) placed between the foundation and the structure, which lengthen the building's natural period and absorb much of the ground motion so that less energy reaches the superstructure.
- **Ductile detailing:** closely spaced steel ties (stirrups) around columns and beams so that reinforced concrete can deform without brittle failure.
- **Lightweight, well-tied roofs** and symmetrical, regular building plans, which avoid dangerous torsional twisting.



Cross-section of an earthquake-resistant building: base isolators decouple the structure from ground motion, while continuous shear walls carry the remaining lateral forces safely down to the foundation.

Retrofitting means strengthening an existing vulnerable building: adding shear walls or steel bracing, jacketing columns with reinforced concrete or steel, tying walls to floors and roofs, and anchoring the structure to its foundation. For Cameroon, where most urban housing in Douala, Yaoundé and Buea was built without any seismic design, cost-effective retrofitting priorities are: enforcing simple codes for new masonry buildings (ring beams, vertical reinforcement at corners, light roofs), strengthening schools and hospitals first, and avoiding construction on loose saturated fills and on steep unstable slopes.

2.4 Community Preparedness and Early Warning

Technology alone cannot save lives; an informed and rehearsed population can.

- **Early warning systems (EWS).** Seismic stations near a fault detect the fast but harmless P-waves and automatically broadcast an alert before the slower, destructive S-waves and surface waves arrive. Depending on distance, this gives seconds to tens of seconds — enough to stop trains, shut gas valves, open fire-station doors and let people take cover ("Drop, Cover, Hold On"). Tsunami warning centres similarly use sea-floor and tide-gauge sensors.
- **Drills.** Regular earthquake and evacuation drills in schools, workplaces and neighbourhoods

turn correct behaviour into reflex. A population that has practised does not panic.

- **Evacuation routes and assembly points.** Routes must be signposted, kept clear of obstructions, lead away from coastal (tsunami) and landslide-prone zones, and converge on open assembly areas where people can be counted and helped.
- **Public education** on securing furniture, storing emergency water and first-aid kits, and knowing safe spots (under a sturdy table, away from windows and heavy objects).

2.5 Post-Event Rapid Damage Assessment

Immediately after an earthquake, authorities must know where the damage is worst in order to direct rescue teams. The standard methodology proceeds in tiers:

1. **Reconnaissance (first hours):** aerial surveys, satellite imagery and drive-through inspections to map the broad pattern of damage and identify cut-off areas.
2. **Rapid visual screening (first days):** trained two-person teams inspect each building from outside (and inside where safe) using a standardised form: construction type, number of storeys, observed damage (none, slight, moderate, severe, collapsed).
3. **Safety tagging:** buildings are posted **green** (safe to occupy), **yellow** (restricted entry — repair needed) or **red** (unsafe — do not enter). This prevents aftershock casualties.
4. **Data compilation:** results are georeferenced and aggregated to produce damage maps, casualty and shelter needs estimates, and priority lists for demolition, shoring and detailed engineering evaluation.

2.6 Quantifying Seismic Risk: A Worked Approach

METHOD Simplified seismic risk index

To compare the relative risk of two districts, use the simplified index:

$$R = \frac{\text{PGA} \times \text{Population density}}{\text{Building resilience factor}}$$

where PGA is the expected peak ground acceleration (in m s^{-2}), population density is in persons per km^2 , and the resilience factor (dimensionless, ≥ 1) summarises building quality — higher for well-engineered, code-compliant stock.

1. Collect or estimate PGA from a seismic hazard map.
2. Obtain population density for the district.
3. Assign a resilience factor (e.g. 1 for poor unreinforced masonry, 3 for code-designed reinforced concrete).
4. Compute R for each district and rank them: the higher R , the higher the relative risk and the higher the priority for mitigation.

WORKED EXAMPLE 2 Comparing seismic risk in two districts

Statement. District A (dense urban quarter on soft alluvium): expected $\text{PGA} = 2.5 \text{ m s}^{-2}$, population density = 20 000 persons/ km^2 , buildings mostly unreinforced masonry (resilience factor = 1). District B (suburb on bedrock): $\text{PGA} = 1.0 \text{ m s}^{-2}$, density = 4 000 persons/ km^2 , mixed building stock (resilience factor = 2). Compute the risk index for each district and comment.

SOLUTION

Solution. For District A:

$$R_A = \frac{2.5 \times 20\,000}{1} = 50\,000.$$

For District B:

$$R_B = \frac{1.0 \times 4\,000}{2} = 2\,000.$$

Hence $R_A/R_B = 25$: District A carries a relative seismic risk about 25 times higher than District B, even though the hazard (PGA) is only 2.5 times greater. The difference is driven by exposure (population density) and vulnerability (low resilience). Mitigation should therefore prioritise District A: enforcement of building codes, retrofitting of schools and hospitals, drills and marked evacuation routes.

WARNING Common mistakes in examinations

- Confusing magnitude (energy at the source, one value per earthquake) with intensity (effects at a site, varies from place to place).
- Saying "a magnitude 6 is twice a magnitude 3": the scale is logarithmic — the energy ratio is $31.6^3 \approx 31\,600$.
- Attributing liquefaction to "mud" or clay: it occurs in saturated loose sands and silts, not in cohesive clays.
- Listing mitigation measures without linking each one to the specific effect it reduces (e.g. base isolation → reduces shaking transmitted to the structure; evacuation routes → reduce exposure to tsunamis and fires).

EXERCISE 2 Applying the concepts

1. A magnitude 5.0 earthquake is followed, one year later, by a magnitude 7.0 event in the same region. Calculate the ratio of the energies released.
2. Explain, in three or four sentences, why a building founded on saturated sand near the

Wouri estuary may suffer more damage than an identical building on bedrock, even at the same distance from the epicentre.

3. A district has expected $\text{PGA} = 3.0 \text{ m s}^{-2}$, population density = 12 000 persons/ km^2 and resilience factor 1.5. Compute its risk index, then state two realistic actions that would lower it, indicating which term of the formula each action changes.

SOLUTION Exercise 1

1. $\frac{E_7}{E_5} = 31.6^{7-5} = 31.6^2 \approx 1000$. The second event releases about one thousand times more energy.
2. Two effects combine: (i) site amplification — soft alluvium amplifies seismic waves, so the ground acceleration at the surface is larger than on bedrock; (ii) liquefaction — shaking raises pore-water pressure in the saturated sand, which loses its strength, so foundations sink, tilt or spread laterally. The building on bedrock experiences weaker shaking on a stable foundation.
3. $R = \frac{3.0 \times 12\,000}{1.5} = 24\,000$. To lower it: (a) retrofit buildings and enforce seismic codes, which raises the resilience factor (denominator); (b) reduce exposure by land-use planning — relocating critical facilities and limiting settlement density in the most hazardous zones — which lowers the population density term. (Strengthening soils or avoiding liquefiable ground effectively reduces the site PGA.)

KEY POINTS Key points of this section

- Primary earthquake effects: **ground shaking** (measured by PGA), **surface rupture** and **liquefaction** of saturated loose soils.
- Secondary effects — **tsunamis**, **landslides**, **fires** and **disease outbreaks** — often exceed primary losses and must never be omitted from a risk assessment.
- The magnitude scale is logarithmic: +1 unit $\approx 31.6\times$ energy.
- Mitigation combines engineering (seismic codes, shear walls, base isolation, retrofitting) and community preparedness (early warning, drills, evacuation routes), followed by structured rapid damage assessment (reconnaissance, screening, green/yellow/red tagging).
- Relative risk grows with hazard (PGA) and exposure (population density) and falls with building resilience: $R = \frac{\text{PGA} \times \text{density}}{\text{resilience}}$.

In the next section we extend this mitigation logic to the other major geohazards — volcanic eruptions, landslides and floods — and examine how land-use planning and hazard mapping complete the engineer's and planner's toolkit.

3 EFFECTS AND MITIGATION OF GEOHAZARDS – PART 2

In the previous section we examined the effects of earthquakes and tsunamis and the first family of mitigation strategies. We now turn to three other geohazards that dominate the Cameroonian risk landscape: **volcanic hazards** along the Cameroon Volcanic Line, **landslides** in the western highlands, and **floods** in the lowlands and coastal plain. For each, we proceed in the same way: observe the phenomenon, understand its mechanism, quantify the danger, and finally distinguish **structural** (engineering) from **non-structural** (planning and social) mitigation.

3.1 Volcanic Hazards

Observation. Mount Cameroon (4095 m), one of Africa's most active volcanoes, erupted in 1999 and 2000, sending lava flows toward the Atlantic coast near Bakingili. Yet lava was not the only threat: ash fell on plantations, and heavy rain on loose ash could have triggered mudflows. A volcano threatens its surroundings through several distinct mechanisms, each with its own speed, reach and lethality.

DEFINITION Volcanic hazard

A **volcanic hazard** is any potentially damaging process associated with a volcano, whether during an eruption (lava flows, pyroclastic density currents, ash fall) or between eruptions (lahars, gas emissions, flank instability).

The four main products and their behaviour.

- **Lava flows.** Streams of molten rock (700 °C to 1200 °C). Basaltic flows, like those of Mount Cameroon, move slowly (typically a few m h⁻¹ to a few tens of km h⁻¹ on steep slopes). They destroy everything by burial and fire but rarely kill, because people can walk away. Mitigation: diversion barriers, cooling with water, and above all zoning.
- **Pyroclastic density currents (PDCs).** Ground-hugging mixtures of hot gas and volcanic fragments travelling at 100–700 km h⁻¹ at temperatures of several hundred degrees. They are the **deadliest** volcanic phenomenon; survival inside a PDC is essentially impossible. Only evacuation before the eruption saves lives.
- **Ash fall (tephra).** Fine fragments ejected into the atmosphere and carried by wind. Ash collapses roofs (wet ash can exceed 200 kg m⁻² of load), contaminates water supplies, damages crops and aircraft engines, and causes respiratory illness.
- **Lahars.** See the definition below — they are the hazard most often underestimated by students.

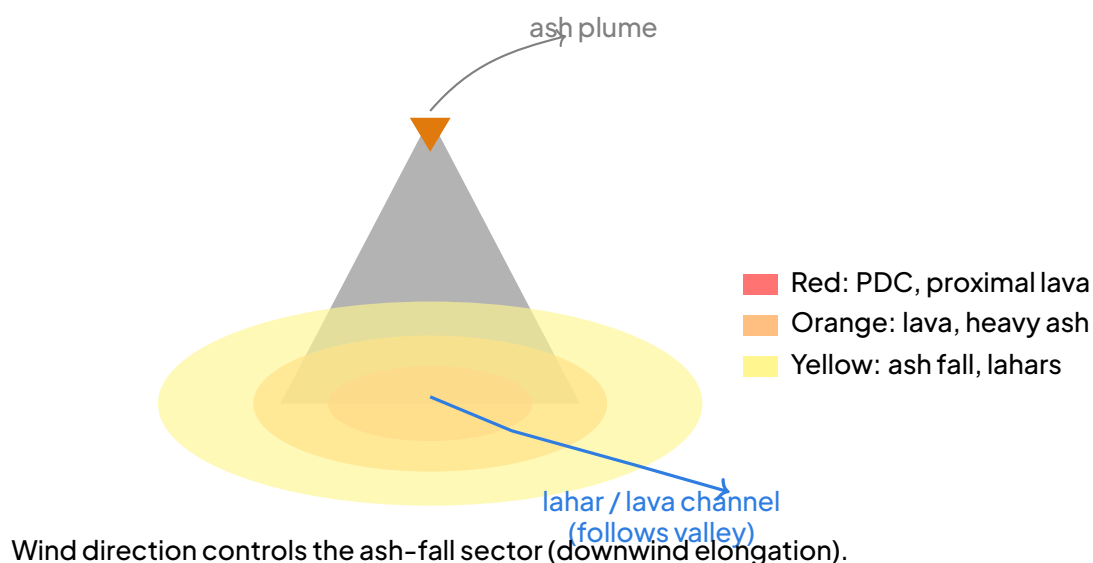
DEFINITION Lahar

A **lahar** is a volcanic mudflow or debris flow: a slurry of volcanic ash, rock fragments and water that rushes down river valleys. It is mobilised by water — from heavy rainfall, melting of summit ice or snow, or the breaching of a crater lake — and can occur long after an eruption, even on a dormant volcano. Lahars behave like wet concrete: fast (up to $\sim 80 \text{ km h}^{-1}$), dense, and able to carry boulders and bury entire settlements.

DID YOU KNOW? Cameroon and volcanic risk

The 1986 Lake Nyos disaster, though a gas (limnic) event rather than an eruption, reminds us that the Cameroon Volcanic Line can kill without any lava at all: about 1700 people died from a cloud of magmatic CO_2 . Degassing pipes now continuously remove CO_2 from the lake bottom — a fine example of structural mitigation.

Hazard zonation. Because each product follows a predictable path (lava and lahars follow valleys; ash follows the wind; PDCs follow topography), geologists draw **volcanic hazard maps** dividing the surroundings of a volcano into zones of decreasing danger.



Schematic volcanic hazard zonation map. Danger decreases outward, but valleys (lahar/lava paths) and the downwind sector (ash) remain hazardous far from the vent.

WARNING Ash fall is not “harmless dust”

A common mistake is to rank hazards only by temperature. Ash fall kills indirectly — roof collapse, water contamination, crop failure — and affects areas hundreds of kilometres downwind, far beyond the red zone. Never omit it from a hazard map.

3.2 Landslides: Triggers and Classification

Observation. After intense rainy-season storms, the steep escarpments of the Bamenda highlands and the slopes around Bafoussam and Dschang frequently fail, burying roads and houses. Why do some slopes stand for centuries and then fail in a single night?

DEFINITION Landslide

A **landslide** is the downslope movement of rock, soil or debris under the direct influence of gravity. It occurs when the driving forces (gravity acting on the slope mass) exceed the resisting forces (cohesion and friction along the potential failure surface).

Triggers (events that push a marginally stable slope to failure):

- **Rainfall** — the dominant trigger in Cameroon. Water increases the weight of the soil and raises **pore-water pressure**, which pushes grains apart and reduces frictional resistance.
- **Earthquakes** — shaking momentarily destroys cohesion and liquefies saturated sands.
- **Human excavation** — road cuts and quarrying remove support at the toe of the slope; overloading at the crest (buildings, waste dumps) adds driving force; deforestation removes root reinforcement.

Classification by movement type (Varnes system, simplified):

- **Falls:** free-fall of rock from a cliff.
- **Topples:** forward rotation of blocks about a pivot.
- **Slides:** movement along a discrete surface — rotational (slumps, curved surface) or translational (planar surface).
- **Flows:** fluid-like movement of saturated material (earthflows, debris flows, mudflows).
- **Creep:** slow, continuous, millimetre-per-year downslope movement.

METHOD Factor of Safety (FoS) for slope stability — planar failure

The **Factor of Safety** quantifies how close a slope is to failure for a planar (translational) slide:

$$\text{FoS} = \frac{\text{resisting forces}}{\text{driving forces}} = \frac{cL + (W \cos \theta - uL) \tan \varphi}{W \sin \theta}$$

where c = cohesion (kPa), L = length of the failure surface, W = weight of the sliding mass,